Jig Dyeing

In this method, the fabric is held on rollers at full width and passed through the dye bath. The fabric is let off from one roller into the dye bath and wound on to the other roller. After one cycle, the process is reversed and the fabric is let off from the second roller to the first. The process is repeated till the desired colour strength is achieved. About 250 kg of fabric can be dyed at a time. This method is preferred for heat-sensitive thermoplastic fabrics that cannot be dyed in rope form as they will form permanent crease or colour streaks in the fabric. This method is not suitable for knitted or stretch fabrics which should not be subjected to tension and also to fabrics where a soft feel is required.

Jet Dyeing

In this method the fabric is dyed in a rope form. It is placed in a heated tube or column where jets of dye solution are thrown on the fabric at high pressure. The fabric also moves along the tube. The dye solution is circulated faster than the fabric while colouring it thoroughly. About 500 kg of fabrics can be jet-dyed at a time. This method of dyeing is suitable for both woven and knitted fabrics made up of wool, cotton, viscose rayon, polyester and its blends.

Pad Dyeing

In pad dyeing, the fabric at full width is passed through a trough containing dye bath. The fabric is then passed between two heavy rollers which force the dye to penetrate in to the fabric and squeeze out the excess dye solution. For the dye to set, the fabric is passed through a heat chamber. Then the dyed fabric is rinsed and dried.

2.3.1.4 Dyeing at Product Stage

This is the last stage of dyeing. When the finished product such as garment is dyed, it is called as product dyeing or garment dyeing. The fabric undergoes pre-treatments such as desizing, bleaching and mercerization before dyeing. The most widely used method for garment dyeing is paddle dyeing. In this method, garments are packed loosely in net bags. Ten to fifty of such bags are kept in large tubs containing dye solution. The dye is agitated by a motor driven paddle which increases the penetration of dye. Garment dyeing is suitable for products such as sweaters, T-shirts, casual clothing and hosiery.

Advantages

- Economical
- No shade variations

Disadvantages

- Colour bleeding and fading.
- Not suitable for products with zippers, sewing, threads, lining etc.

2.3.2 Dyeing of Blended Fabrics

When a fabric is made of one type of fibre/ yarn, dyeing process is simple but when it is composed of more than one kind of yarn as in a blend, dyeing process is complex. Special procedures are carried out where different dyes that are suitable for each fibre need to be selected.

2.3.2.1 Union Dyeing

In this method, blended fabrics are dyed to the same shade to get a solid-coloured fabric. Fabrics can be dyed by one or two bath process. In one bath process, dyes appropriate for each fibre is added in same dye bath. In two bath process, dyes are added separately to each fibre subsequent to one another.

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